

Problem Response



Multiple-Choice Questions

Exact answer matching

Binary Score: 0/1



Open-Ended QA Questions

Adaptive Process-Based Scoring Pipeline

Phase 1: Logic Alignment & Routing



Selector LLM compares model response with reference solution
Determine whether reasoning is:

Aligned

Divergent



*Supports fairness
for alternative valid
solution paths*

Phase 2A: Granular Step-Based Verification

Decomposition

Reference solution $\rightarrow$ atomic logical steps
Assign point values (sum=10)

Grading Rubric

Explicit requirements for logic and intermediate results:
Same rubric used across model evaluations

Step-by-step Evaluation

Evaluator maps student response to rubric
Award verified partial credit only

Normalized Score

Normalize to [0,1]

Difficulty-aware Rubric

Easy

Terminal proof process

Medium

Terminal proof process

Hard

Full independent reasoning chain



*Penalizes
unjustified
logical leaps*

Phase 2B: Independent Path Verification & Scoring

Parse Generated Reasoning

Construct a custom step framework

Strict Validation

Check theorem/tool preconditions
Penalize unstated premises or unmet conditions

Step-wise Assessment

Logical Coherence
(S_{logic})

Technical Precision
(S_{tech})

Global scores = average across steps

Final Aggregation

Proof tasks: $S_{final} = w_1 \cdot S_{logic} + w_2 \cdot S_{tech}$

Quantitative derivation tasks:

$S_{final} = w_1 \cdot S_{logic} + w_2 \cdot S_{tech} + w_3 \cdot S_{result}$

Normalize to [0,1]

Final Score Stabilization

Run the full evaluation 3 independent times
Final score = average of 3 trials

*Balances cost and
LLM randomness*